# build_reports

## User Manual

Flow Metrics for Agile Teams - Setup and Usage

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situation-report - github.com/Jaegerfeld/situation-report

For non-technical users — Version 0.5.0

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# 1 What is build_reports?

build_reports is a tool that automatically creates meaningful charts about the progress and efficiency of your agile team. As input it uses the data that the **transform_data** module has exported from your issue tracker (e.g. Jira). build_reports reads these files and calculates several **flow metrics** — graphical analyses showing how fast and how much your team delivers.

The program has a simple graphical user interface (GUI): no programming knowledge required. At the click of a button, charts are displayed in the browser or saved as a PDF file.

**Metrics overview**
- **Flow Time / Cycle Time**: How long does it take for an issue to be completed?
- **Flow Velocity / Throughput**: How many issues does the team close per week?
- **Flow Load / WIP**: How many issues are in progress simultaneously?
- **Cumulative Flow Diagram**: How does the inventory develop over time?
- **Flow Distribution**: How are issues distributed across types, stages and cycle times?
- **Process Flow**: Which status paths do issues take? Where do rework and loops occur?

# 2 Prerequisites and Installation

## 2.1 What needs to be installed?

For build_reports to work, **Python 3.11 or newer** must be installed on the computer. Additionally, several Python packages must be present. Whoever set up the program for you should have already handled this.

## 2.2 Starting the program

There are two ways to start build_reports:

- **Double-click** on the file **build_reports_gui.pyw** in the project folder — opens the graphical interface without a console window.
- **Terminal / command prompt**: Navigate to the project directory and type `python -m build_reports`.

> **Tip:** If you use the GUI regularly, you can create a shortcut to **build_reports_gui.pyw** on your desktop.

# 3 Input Files

build_reports requires one or two Excel files produced by the **transform_data** module. These files must not be edited manually — the structure must exactly match the expected format.

## 3.1 IssueTimes.xlsx (required)

This file contains all issues (tickets) with their time data and current processing status. It is required for all metrics except the Cumulative Flow Diagram.

| Column | Meaning |
|---|---|
| Project | Project key (e.g. ARTA) |
| Key | Issue key (e.g. ARTA-123) |
| Issuetype | Type of issue (e.g. Feature, Bug, Story) |
| Status | Current status (e.g. In Progress, Done) |
| Created | Creation date of the issue |
| First Date | Date on which the issue was first actively worked on |
| Closed Date | Completion date (empty = still open) |
| Resolution | Resolution type (e.g. Fixed, Duplicate) |
| Stage columns | One column per workflow stage with minutes spent in that stage |

## 3.2 CFD.xlsx (optional, for Cumulative Flow Diagram)

This file contains daily entry counts: how many issues **entered** a given stage on each day (not snapshots). build_reports accumulates these values into a running total. It is only needed if the Cumulative Flow Diagram is to be calculated.

| Column | Meaning |
|---|---|
| Day | Date (YYYY-MM-DD) |
| Stage columns | One column per stage with the number of new entries on that day |

## 3.3 PI configuration file (optional, for Flow Velocity)

With an optional JSON configuration file you can define your own PI intervals (Program Increments) for the Flow Velocity bar chart. Without this file, calendar quarters are used automatically.

**Example (date mode):**

```
{ "mode": "date",
  "intervals": [
    {"name": "PI 2025.1", "from": "2025-01-06", "to": "2025-04-04"},
    {"name": "PI 2025.2", "from": "2025-04-05", "to": "2025-07-04"},
    {"name": "PI 2025.3", "from": "2025-07-05", "to": "2025-10-03"},
```

```
    {"name": "PI 2025.4", "from": "2025-10-04", "to": "2026-01-02"}
  ]
}
```

The file must have a **.json** extension. Copy the provided example file **pi_config_example.json** and adjust the dates and names to match your PI schedule. The format must be preserved.

> **Note:** Always use date format **YYYY-MM-DD** (year-month-day). Example: January 6, 2025 = 2025-01-06.

## 3.4 Transitions.xlsx (optional, for Process Flow)

This file contains all status transitions per issue in chronological order. It is produced by the **transform_data** module and is required exclusively for the **Process Flow metric**. All other metrics can be calculated without this file.

| Column | Meaning |
|---|---|
| Key | Issue key (e.g. ARTA-123) |
| Transition | Target status after the transition (e.g. 'In Analysis') |
| Timestamp | Timestamp of the transition (DD.MM.YYYY HH:MM:SS) |

> **Note:** Transitions.xlsx and IssueTimes.xlsx must come from the same transform_data export run so that issue keys match.

# 4 The Graphical User Interface (GUI)

After starting, the main window opens. It consists of three areas: the **file area** (top), the **filter area** (middle) and the **action area** (bottom) with the log window.

## 4.1 Loading files

First load the required files:

- **IssueTimes** — Click the folder button to the right of the field and select the **IssueTimes.xlsx** file. After loading, available projects and issue types appear automatically in the log.
- **CFD (optional)** — Select the **CFD.xlsx** file if you need the Cumulative Flow Diagram.
- **Workflow (optional)** — Select the workflow text file from transform_data. It contains **<First>** and **<Closed>** markers that determine which stage boundaries the CFD trend lines mark.
- **PI config (optional)** — Select your JSON configuration file for custom PI intervals. Leave the field empty to use calendar quarters.
- **Transitions (optional)** — Select the **Transitions.xlsx** file from transform_data. Only required if the Process Flow metric is to be calculated.

> **Tip:** Hovering over an input field shows a tooltip explaining what the field is used for.

## 4.2 Setting filters

Filters restrict which issues are included in the analysis:

| Filter / Exclusion | Description |
|---|---|
| From / To | Only consider issues closed within this date range. Format: YYYY-MM-DD. The calendar button opens |
| Last 365 days | Automatically sets From and To to the last 365 days up to today. |
| Projects | Only analyse specific projects. Separate multiple projects with a comma, e.g. ARTA, ARTB. The selecti |
| Issue types | Only analyse specific issue types, e.g. Feature, Bug. Leave empty = all types. The selection button sho |
| Exclude: Status | Completely remove issues with certain Jira statuses from all metrics, e.g. 'Canceled'. The selection butt |
| Exclude: Resolution | Exclude issues with certain resolution types, e.g. 'Won't Do' or 'Duplicate'. The selection button shows a |
| Exclude zero-day issues | Checkbox: issues whose cycle time (First to Closed Date) is shorter than the configured threshold are re |

## 4.3 Selecting metrics and CT method

Use the checkboxes to select which metrics should be calculated. The **All** and **None** buttons set or clear all checkboxes at once.

The **CT method** determines how cycle time is calculated — relevant only for the Flow Time metric:

| Method | Calculation |
|---|---|
| Method A (default) | Difference in calendar days between First Date and Closed Date. Simple and straightforward. |

| Method | Calculation |
|---|---|
| Method B | Sum of minutes in the individual workflow stages (last stage excluded), divided by 1440. Measures onl |

## 4.4 Creating a report

You have two options:

- **Show in browser** — All charts are opened in your default browser. The charts are fully interactive there: zoom in, inspect data points via hover tooltip, and toggle individual categories in the legend.
- **Export reports** — All charts are exported to a multi-page PDF file. A save dialog asks for file name and location. In addition to the PDF, two Excel files are automatically created: a report Excel with all issues, status groups and cycle times, and — if zero-day issues exist — a separate file for those issues.

> **Note:** While calculations are running, the interface is briefly locked. Progress is shown in the log window. Please do not close or click until the log shows the completion message.

## 4.5 Templates — saving and loading configuration

In the **Templates** menu you can save all current settings (file paths, filters, metric selection, CT method, terminology) as a JSON file and reload them later — no need to fill in all fields every time.

- **Save...** — Choose a location and a name for the configuration file (e.g. myTeam_QuarterlyReport.json).
- **Load...** — Open a saved configuration file. All fields are filled in automatically. If a saved file can no longer be found, a note appears in the log.

## 4.6 Language and terminology

In the **Options** menu you can switch language and terminology:

- **Language** — Switch between German and English. All labels, tooltips and menu items are updated immediately.
- **Terminology** — Switch between **SAFe** and **Global**. In SAFe mode the metrics are called e.g. 'Flow Time', in Global mode 'Cycle Time'. This switch affects only the labels, not the calculations.

# 5 Metrics Overview

This section explains each metric in plain language: what it measures, what the charts show and how to interpret the results. The example charts are based on a sample dataset.

## 5.1 Flow Time / Cycle Time

**What is measured?** The cycle time — i.e. the number of days an issue takes from first work to completion. Shorter is better.

### *Chart 1: Box plot (distribution)*

The box plot shows at a glance how cycle times are distributed. The chart header contains the key statistics:

| Statistic | Meaning |
|---|---|
| Min / Max | Shortest and longest measured cycle time |
| Q1 / Q3 | 25% / 75% of issues fall below this value |
| Median | The middle cycle time — 50% of issues fall below it |
| Mean | Average cycle time (can be skewed by outliers) |
| 90d CT% | Share of issues with cycle time <= 90 days (Service Level Expectation) |
| P85 / P95 | 85% / 95% of issues were completed within this time |
| Std dev | Standard deviation — how much do the values vary? |
| CV | Coefficient of variation — relative spread (smaller = more stable process) |
| Zero-Day | Number of issues with cycle time 0 (excluded from the analysis) |

*Red dot in the box plot = statistical outlier. In the browser you can read the issue key via hover tooltip.*

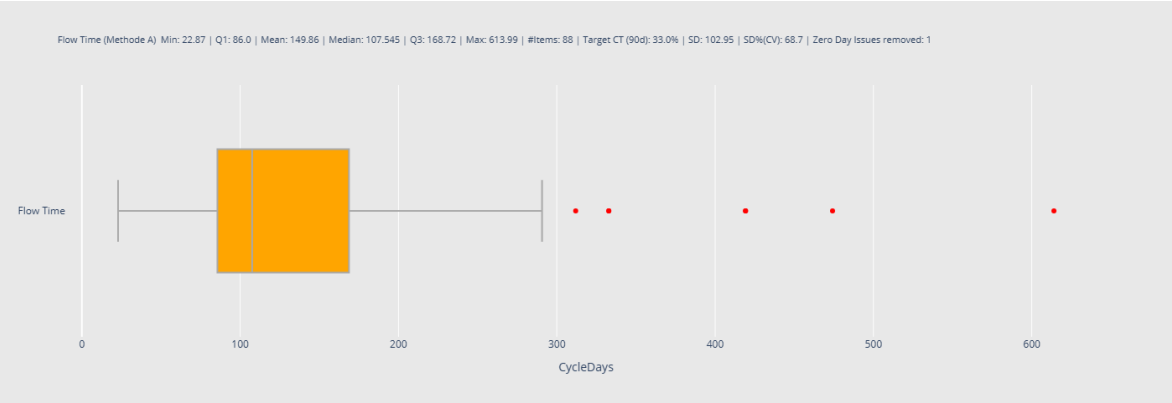


*Fig. 1: Box plot of cycle times — distribution, quartiles and statistics header.*

### *Chart 2: Scatter plot (trend over time)*

Each dot is a completed issue. The x-axis shows the completion date, the y-axis the cycle time in days. Colours and reference lines aid interpretation:

| Element | Meaning |
|---|---|
| Blue dot | Normal issues (below the 85th percentile) |
| Orange dot | Slow issues (between 85th and 95th percentile) |
| Red dot | Very slow issues (above the 95th percentile) |
| Blue curve | LOESS trend line — shows the cycle time trend over time |
| Red line | Median reference line |
| Green line | 85th percentile reference line |
| Cyan line | 95th percentile reference line |

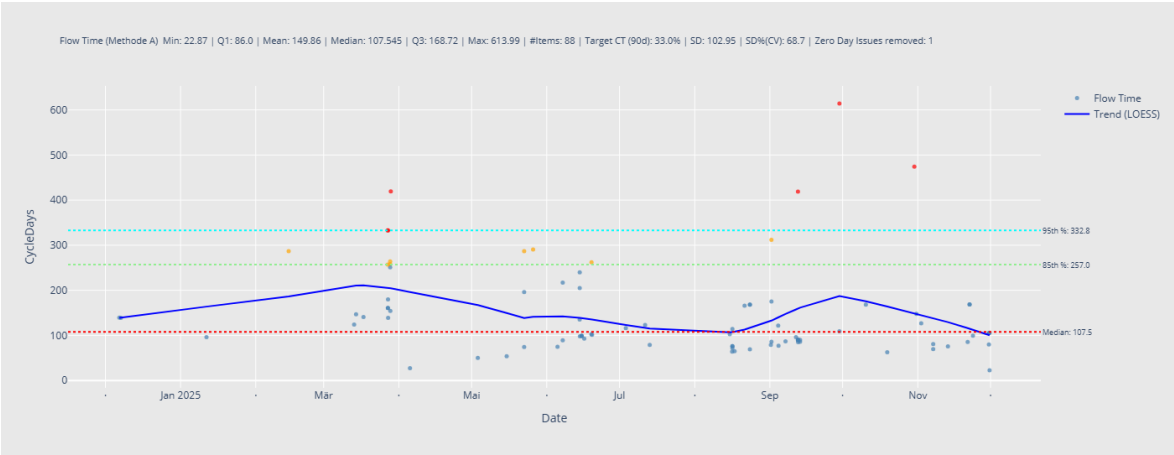


*Fig. 2: Scatter plot — cycle time per completion date with LOESS trend line and reference lines.*

**Interpretation:** If the LOESS trend line rises to the right, issues are getting slower over time. A flat line signals a stable process. Many red and orange dots indicate frequent bottlenecks.

## 5.2 Flow Velocity / Throughput

**What is measured?** Throughput — i.e. how many issues the team closes per week or per PI. Consistently high velocity indicates a delivery-capable team.

| Chart | Shows |
|---|---|
| Daily frequency (histogram) | How often exactly 1, 2, 3 ... issues are closed on a single day. Shows typical daily output. |
| Weekly trend (line chart) | Number of issues closed per week over the entire period. Fluctuations and trends become immedia |
| PI trend (bar chart) | Number of issues closed per PI (Program Increment) or quarter. The red line shows the average. E |

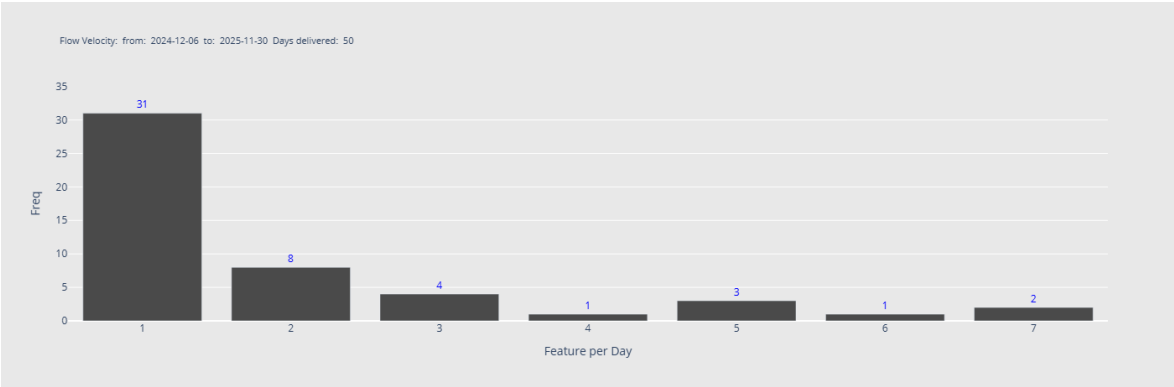


*Fig. 3: Daily frequency — frequency of daily closure counts.*

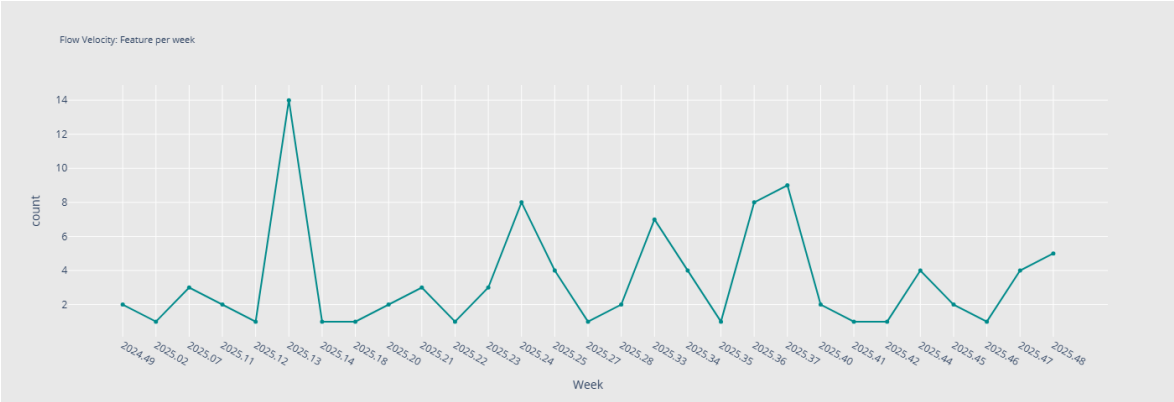


*Fig. 4: Weekly trend — issues closed per calendar week.*

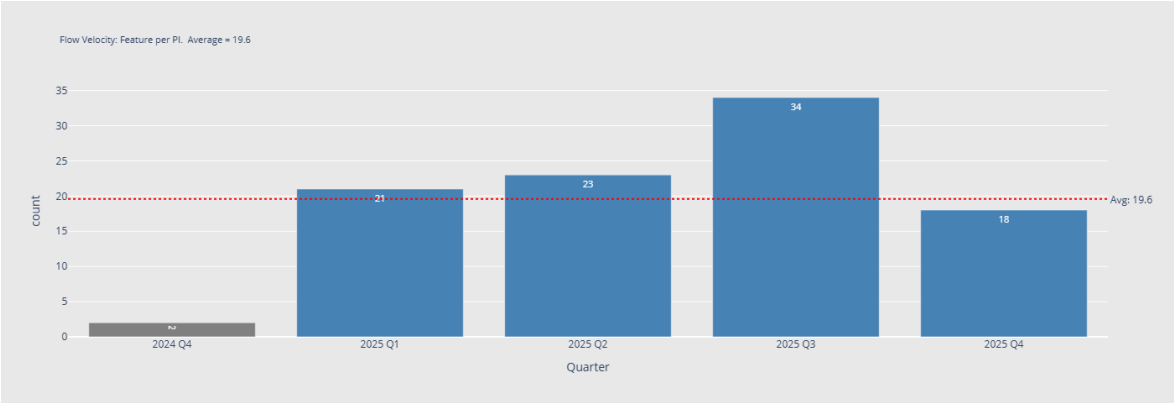


*Fig. 5: PI trend — issues closed per PI or quarter with average line.*

## 5.3 Flow Load / WIP (Work in Progress)

**What is measured?** How many issues are simultaneously in progress and how old they already are. Too many parallel issues slow down delivery (the more WIP, the longer the cycle time).

The chart shows a grouped box plot: each stage gets a box showing the age (in days) of the issues currently there. Individual dots represent individual issues — in the browser you see the issue key on hover.

Dashed reference lines from closed issues (median, 85th percentile, 95th percentile) provide orientation: issues already above the 95th percentile of closed issues are significantly delayed.

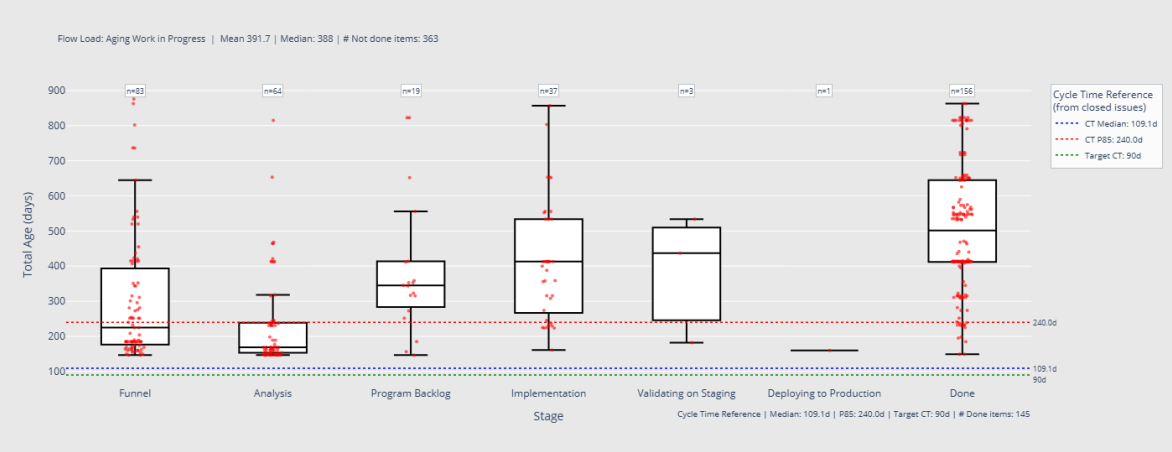


*Fig. 6: Flow Load — age of open issues per stage with reference lines from closed issues.*

## 5.4 Cumulative Flow Diagram (CFD)

**What is measured?** How many issues in total have entered each stage — cumulated over time, broken down by workflow stage. A well-functioning system shows parallel, evenly rising bands without swelling in individual stages.

The chart is a stacked area diagram: each coloured layer corresponds to a stage. The first stage is at the top, the last (Done/Closed) at the bottom. The chart always starts at 0 — regardless of the selected start date. Two black trend lines show:

- **Upper line (inflow):** Runs along the visual top edge of the <First> stage (system entry). Without a workflow file: first stage.
- **Lower line (outflow):** Runs along the visual top edge of the <Closed> stage (system completion). Without a workflow file: last stage.

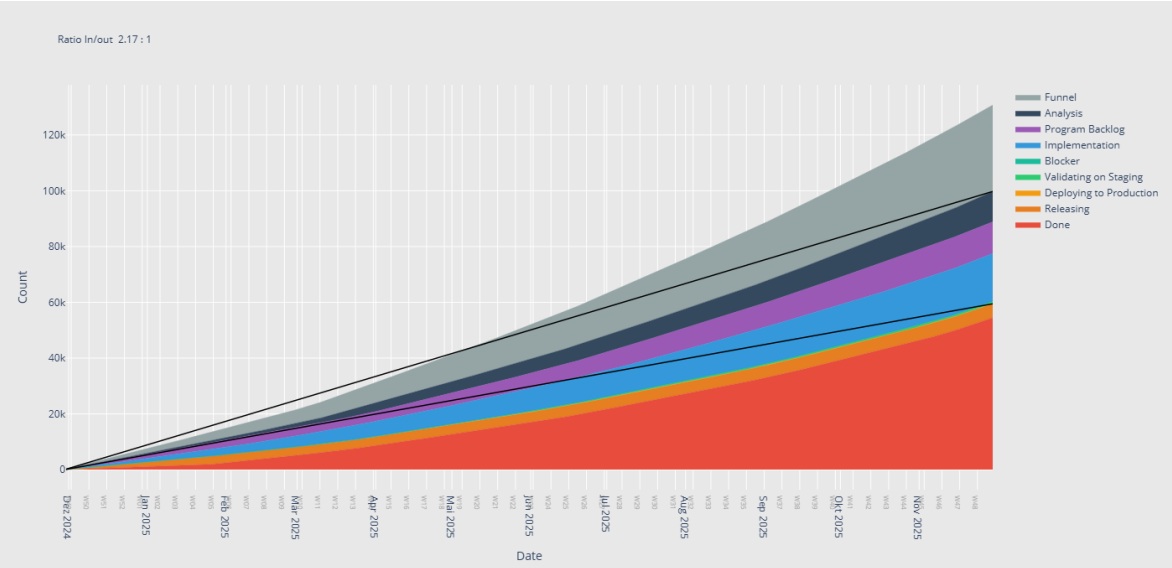


*Fig. 7: Cumulative Flow Diagram — cumulative entries per stage with inflow and outflow trend lines.*

The **In/Out ratio** in the chart title (e.g. 'Ratio In/out 1.80 : 1') shows whether more is coming in than being completed. A value of 1.0 means a balanced system; values significantly above 1.0 indicate a

growing backlog.

The x-axis shows month boundaries with large labels (e.g. 'Jan 2025') and ISO calendar weeks with small grey labels (e.g. 'W03'), so that labels do not overlap.

> **Note:** The CFD requires the optional CFD.xlsx file. Without this file the CFD metric cannot be calculated.

## 5.5 Flow Distribution

**What is measured?** The composition of all issues by type, dominant stage and average cycle time. Shows at a glance which issue types dominate, where issues spend most time, and which types are processed fastest or slowest.

The chart consists of three sub-charts side by side:

| Chart | What is shown? |
|---|---|
| By Issue Type (donut) | Count and percentage share of issues per issue type. All issues are included. |
| Stage Prominence (donut) | For each issue the stage in which it spent the longest time is identified. The chart counts how often |
| Avg Cycle Time by Type (bar) | Average cycle time in days per issue type (Method A: Closed Date - First Date). Only issues with b |

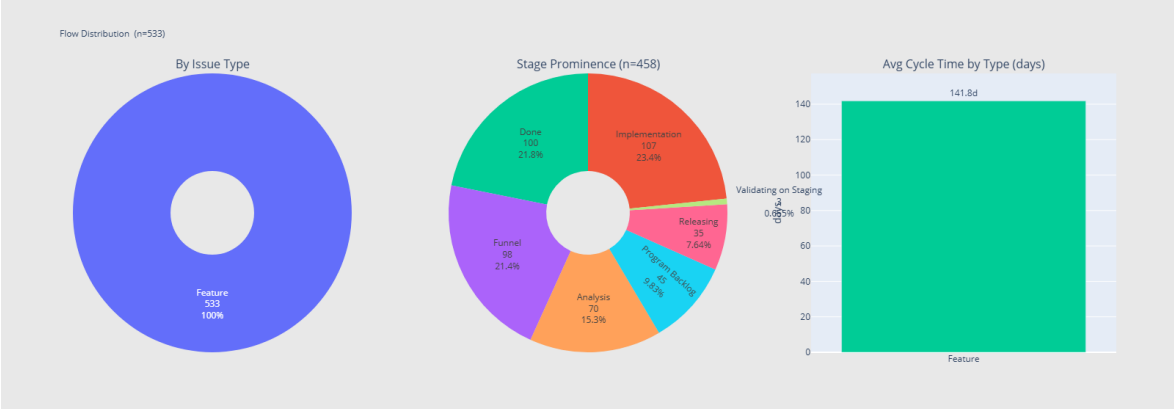


*Fig. 8: Flow Distribution — issue type distribution, Stage Prominence and avg cycle time per type.*

> **Interpretation Stage Prominence:** If a stage dominates particularly often, issues linger there disproportionately long — a potential bottleneck in the workflow. Closed issues are included, but their terminal Done stage is hidden, so actual processing bottlenecks remain visible.

## 5.6 Process Flow

**What is measured?** All status transitions of issues are visualised as a directed graph: nodes = statuses, arrows = transitions. Arrow thickness is proportional to the frequency of the transition. This makes it immediately clear which paths issues take through the workflow, how often rework occurs, and where issues get stuck in loops.

This metric requires the optional **Transitions.xlsx** file (from transform_data). Without this file a warning appears in the log.

| Element | Meaning |
|---|---|
| Blue arrow | Forward transition — issue moves forward in the workflow. |
| Red arrow | Backward transition (rework) — issue returns to an earlier stage. |
| Orange arc | Self-loop — issue stays in the same status (e.g. stage traversed again). |
| Arrow width | The thicker the arrow, the more frequent this transition. |
| Number on arrow | Absolute count of this transition across all issues. |
| Node | Dark blue circle with status name. Order: workflow stages first (clockwise), then additional statuses alp |

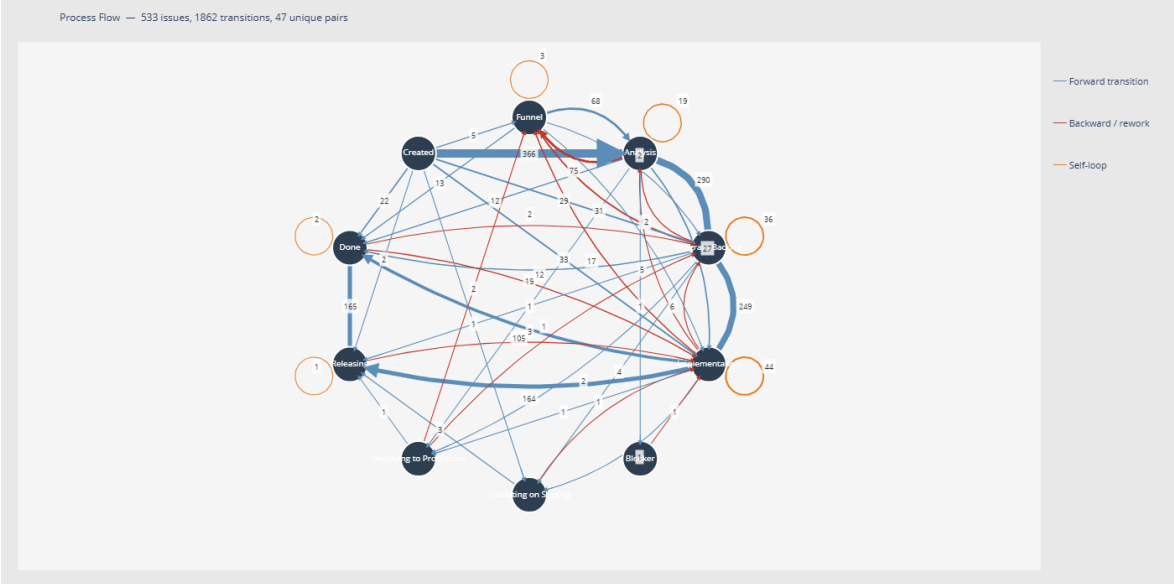


*Fig. 9: Process Flow — directed graph of all status transitions with edge width and colour coding.*

**Interpretation:** Many red arrows mean frequent rework — a sign of quality issues or unclear requirements. Thick blue arrows show the main workflow path. Self-loops (orange) occur when an issue is set to the same status multiple times.

# 6 PDF Export

The PDF export creates a multi-page PDF file with all selected charts. Each chart appears on its own page.

| Step | Action |
|---|---|
| 1 | Load files and set filters (as described in Chapter 4). |
| 2 | Select desired metrics via checkboxes. |
| 3 | Click 'Export reports'. |
| 4 | In the save dialog, choose a file name and location and confirm. |
| 5 | The program calculates and exports; progress appears in the log. |
| 6 | After completion, the PDF and report Excel are available at the chosen location. |

## 6.1 Automatic report Excel

With every PDF export an Excel file with the same name is automatically created (e.g. report.xlsx next to report.pdf). This file contains all filtered issues in IssueTimes format, supplemented by three columns:

| Column | Content |
|---|---|
| Status Group | Status group of the issue: 'To Do' (not yet started), 'In Progress' (being worked on) or 'Done' (co |
| Cycle Time (First->Closed) | Cycle time in calendar days from First Date to Closed Date (Method A). Empty if either date is r |
| Cycle Time B (days in Status) | Sum of minutes in all workflow stages except the last, divided by 1440 (Method B). Empty if eith |

**Zero-day issues:** Two mechanisms operate independently:
1. **Exclusion filter (before calculation):** If the checkbox 'Exclude zero-day issues' is active, issues with a cycle time below the configured threshold (default: 5 minutes) are completely removed from all metrics.
2. **Within the Flow Time metric:** Issues with a cycle time of 0 days (same calendar day) are reported separately and not included in the statistics.
In both cases a separate Excel file is created (e.g. report_zero_day_issues.xlsx in the same folder).

# 7 Frequently Asked Questions

### *Q: The charts do not appear in the browser.*

A: Check whether a default browser is configured. Alternatively try the PDF export. Make sure the IssueTimes file was loaded correctly (check the log).

### *Q: The PDF export takes very long or fails.*

A: Rendering charts as PDF requires the Kaleido package. If this has not yet been set up, contact your technical contact.

### *Q: The log shows 'Stage only in IssueTimes' or 'Stage only in CFD'.*

A: The stage columns in IssueTimes.xlsx and CFD.xlsx do not match. This is a warning that does not stop the analysis, but indicates that the files come from different workflow versions.

### *Q: How can I analyse only a specific project?*

A: Enter the desired project key in the 'Projects' field (e.g. ARTA). Separate multiple projects with a comma. Alternatively: use the selection button for a list of all available projects.

### *Q: The Cumulative Flow Diagram does not appear.*

A: The CFD metric requires a CFD.xlsx file. Load it in the 'CFD (optional)' field.

### *Q: Process Flow shows 'No transition data available'.*

A: The Process Flow metric requires a Transitions.xlsx file from transform_data. Load it in the 'Transitions (optional)' field. Make sure the file comes from the same export run as the IssueTimes.xlsx.

### *Q: What is the difference between PI intervals and quarters?*

A: By default, calendar quarters (Q1-Q4) are used as time intervals. With a PI configuration file you can define your own intervals that match your actual PIs — for example if your PI starts on January 6 instead of January 1.

### *Q: How do I save my settings?*

A: Use the menu 'Templates' -> 'Save...' to save all current settings in a JSON file. Next time: 'Templates' -> 'Load...'. Exclusion settings can additionally be stored permanently under 'Templates' -> 'Save exclusions as default'.

### *Q: An issue appears in the metrics even though it was never really worked on.*

A: This happens when an issue was manually clicked through all workflow stages within seconds — without any real development work. Enable the checkbox 'Exclude zero-day issues' under 'Exclusions' in the GUI (threshold e.g. 5 minutes). The issue is then completely removed from all metrics and documented in a separate Excel file.

### *Q: Can I present the results without a computer?*

A: Yes: first export a PDF report. The PDF file contains all charts and can be opened on any device. For interactive presentations, the browser view is recommended.

# 8 Glossary

| Term | Explanation |
|---|---|
| Closed Date | The date on which an issue was completed. |
| Cycle Time | Alternative term for Flow Time (Global terminology). |
| First Date | Date of the first active work on an issue. |
| Flow Load | Number of issues currently in progress (SAFe term). |
| Flow Time | Cycle time from first work to completion. |
| Flow Velocity | Number of issues completed per time period (SAFe term). |
| Issue | A ticket in the issue tracker (e.g. a Jira card). |
| Issue type | Category of an issue, e.g. Feature, Bug, Story, Task. |
| IssueTimes | The Excel file with all issues produced by transform_data. |
| JSON | Simple text format for configuration files. |
| LOESS | Statistical smoothing method for trend lines. |
| P85 / P95 | 85th / 95th percentile of cycle times. |
| PI | Program Increment — a fixed planning and delivery period. |
| Process Flow | Directed graph of all status transitions. Shows main paths, rework, and loops in the workflow. |
| Resolution | Resolution type of an issue, e.g. 'Done', 'Won't Do', 'Duplicate'. |
| SAFe | Scaled Agile Framework — a framework for agile scaling. |
| Stage | A step in the workflow, e.g. Analysis, Implementation, Done. |
| Template | Saved configuration file with all settings. |
| Throughput | Alternative term for Flow Velocity (Global terminology). |
| Transitions | Record of every status change per issue. Exported by transform_data as Transitions.xlsx. |
| WIP | Work in Progress — issues that are currently being worked on. |
| Zero-day issue | An issue whose cycle time (First to Closed Date) is so short that it does not represent real processing |